Experiment 20 Experiment to understand K-means clustering algorithm for a given data set.

Description

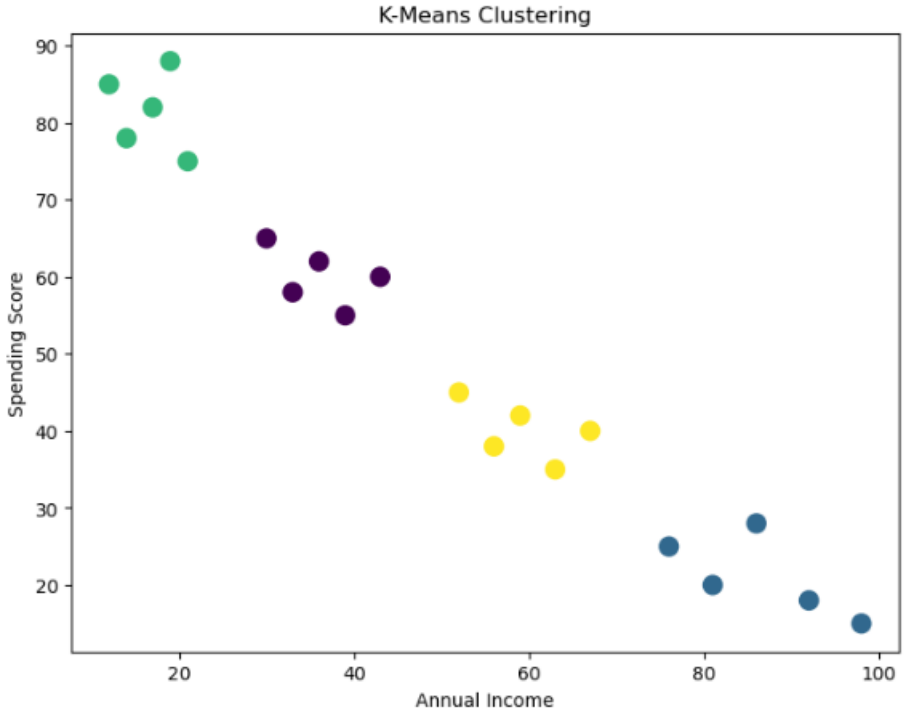
The program demonstrates the K-Means clustering algorithm by grouping customers based on their annual income and spending score. The data is divided into five clusters and visualized using a scatter plot.

Program:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
# New dataset
data = {
    "Annual_Income": [12, 14, 17, 19, 21, 30, 33, 36, 39, 43,
                      52, 56, 59, 63, 67, 76, 81, 86, 92, 98],
    "Spending_Score": [85, 78, 82, 88, 75, 65, 58, 62, 55, 60,
                       45, 38, 42, 35, 40, 25, 20, 28, 18, 15]
}
df = pd.DataFrame(data)
print("Dataset:")
print(df)
print("\nDataset Information:")
df.info()
print("\nFirst 5 Rows:")
print(df.head())
# Select features
features = df[["Annual_Income", "Spending_Score"]].values
# Create K-Means model
model = KMeans(n_clusters=4, random_state=42, n_init=10)
# Train the model
model.fit(features)
# Add cluster labels
```

```
df["Cluster"] = model.predict(features)
print("\nClustered Dataset:")
print(df)
# Plot clusters
plt.figure(figsize=(8, 6))
plt.scatter(
    df["Annual_Income"],
    df["Spending_Score"],
    c=df["Cluster"],
    s=100
)
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.title("K-Means Clustering")
plt.grid(False)
plt.show()
```

Output:



```
Dataset:
    Annual_Income  Spending_Score
0              12              85
1              14              78
2              17              82
3              19              88
4              21              75
5              30              65
6              33              58
7              36              62
8              39              55
9              43              60
10             52              45
11             56              38
12             59              42
13             63              35
14             67              40
15             76              25
16             81              20
17             86              28
18             92              18
19             98              15

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 2 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Annual_Income   20 non-null     int64
 1   Spending_Score  20 non-null     int64
dtypes: int64(2)
memory usage: 392.0 bytes

First 5 Rows:
   Annual_Income  Spending_Score
0             12              85
1             14              78
2             17              82
3             19              88
4             21              75
/lib/python3.14/site-packages/threadpoolctl.py:1135: RuntimeWarning: JsProxy.as_object_map() is deprecated. Use as_py_json() instead.
  for filepath in LDSO.loadedLibsByName.as_object_map():

Clustered Dataset:
    Annual_Income  Spending_Score  Cluster
0              12              85        2
1              14              78        2
2              17              82        2
3              19              88        2
4              21              75        2
5              30              65        0
6              33              58        0
7              36              62        0
8              39              55        0
9              43              60        0
10             52              45        3
11             56              38        3
12             59              42        3
13             63              35        3
14             67              40        3
15             76              25        1
16             81              20        1
17             86              28        1
18             92              18        1
19             98              15        1
```